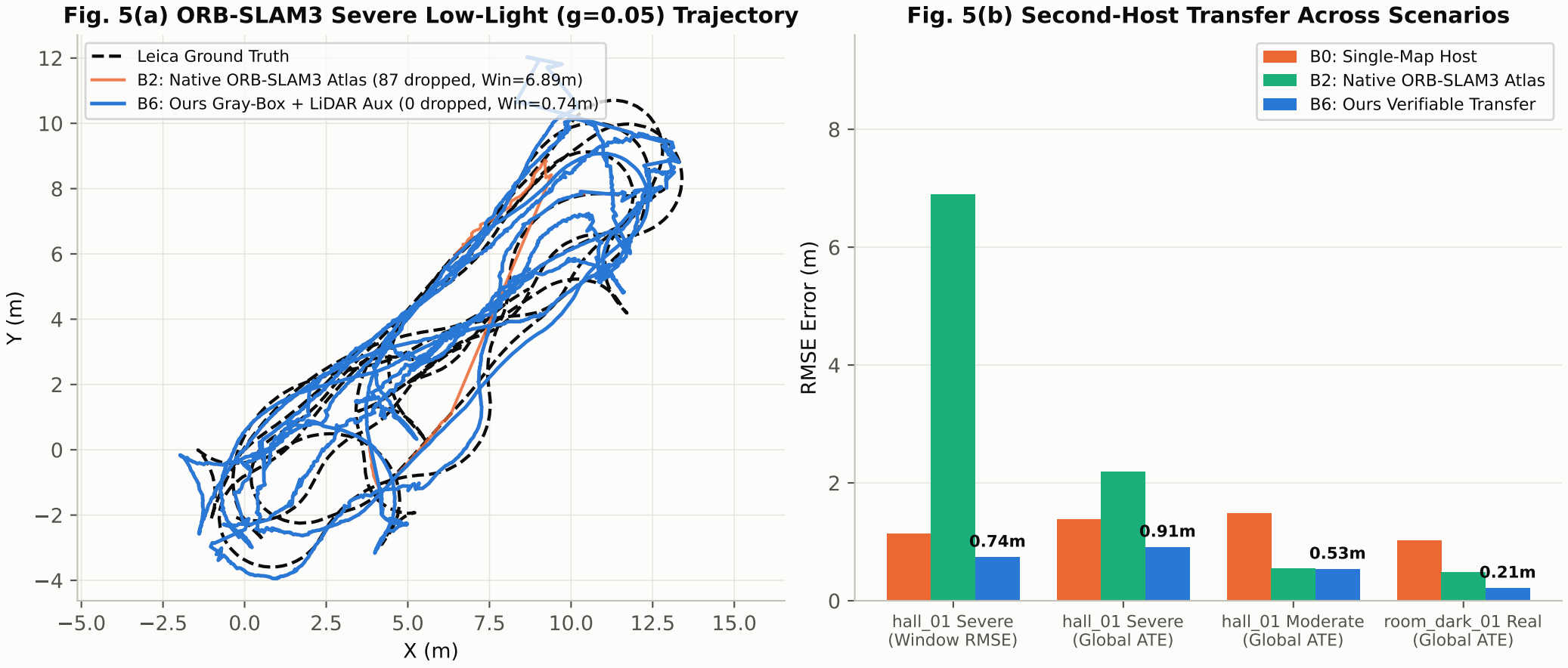

Fig. 5(a) ORB-SLAM3 Severe Low-Light (g=0.05) Trajectory
Leica Ground Truth
B2: Native ORB-SLAM3 Atlas (87 dropped, Win=6.89m)
B6: Ours Gray-Box + LiDAR Aux (0 dropped, Win=0.74m)
Y (m)
X (m)
Fig. 5(b) Second-Host Transfer Across Scenarios
B0: Single-Map Host
B2: Native ORB-SLAM3 Atlas
B6: Ours Verifiable Transfer
RMSE Error (m)
0.74m
0.91m
0.53m
0.21m
hall_01 Severe (Window RMSE)
hall_01 Severe (Global ATE)
hall_01 Moderate (Global ATE)
room_dark_01 Real (Global ATE)